

MODULE 5 - Restrictions on Use of Hazardous Substances (ROHS)

Hazardous Substances in E-Waste

1. Introduction

E-waste, or electronic waste, refers to discarded electrical and electronic equipment (EEE) such as computers, mobile phones, televisions, refrigerators, and other electronic gadgets. Although e-waste often contains valuable materials like gold, copper, and aluminium, it also contains a variety of hazardous substances that pose serious environmental and health risks if not handled properly.

2. Major Hazardous Substances in E-Waste

Category	Substance	Common Sources	Hazards
Heavy Metals	Lead (Pb)	CRT monitors, solder in PCBs, batteries	Neurotoxic; affects brain development, kidneys, and blood system
	Mercury (Hg)	LCD backlights, switches, thermostats	Damages brain and kidneys; bioaccumulates in food chains
	Cadmium (Cd)	Rechargeable Ni-Cd batteries, semiconductors, resistors	Carcinogenic; damages kidneys and bones
	Chromium (Cr VI)	Corrosion protection coatings, metal housings	Carcinogenic; causes skin and respiratory problems
	Beryllium (Be)	Motherboards, connectors	Causes lung disease (berylliosis) when inhaled
Halogenated Compounds	Polybrominated Flame Retardants (PBDEs, PBBs)	Plastic casings, circuit boards, cables	Persistent organic pollutants; endocrine disruptors

Category	Substance	Common Sources	Hazards
	Polychlorinated Biphenyls (PCBs)	Capacitors, transformers (older devices)	Toxic; affects immune, reproductive, and nervous systems
Gases and Others	Chlorofluorocarbons (CFCs)	Refrigerators, air conditioners	Deplete ozone layer; contribute to global warming
	Polyvinyl Chloride (PVC)	Cable insulation, casings	Releases dioxins and furans when burned; carcinogenic
	Arsenic (As)	Semiconductors, LCD panels	Toxic to liver, skin, and nervous system

3. Sources of Hazardous Substances in E-Waste

1. **Printed Circuit Boards (PCBs)** – Contain lead solder, brominated flame retardants, and beryllium.
2. **Cathode Ray Tubes (CRTs)** – High levels of lead in glass and phosphor coatings.
3. **Batteries** – Cadmium, lead, mercury, and lithium compounds.
4. **Plastics and Casings** – Contain PVC and brominated flame retardants.
5. **Capacitors and Transformers** – May have PCBs (in older equipment).
6. **LCD and Plasma Displays** – Contain mercury in backlights.
7. **Refrigeration Units** – Use CFCs and HCFCs as coolants.

4. Environmental Impact of Hazardous Substances

1. Soil Contamination:

Improper disposal (open dumping or landfilling) causes leaching of heavy metals into soil, reducing fertility and entering food chains.

2. Water Pollution:

Leachate containing lead, cadmium, and mercury can contaminate groundwater and surface water.

3. Air Pollution:

Open burning of cables and plastics releases toxic fumes (dioxins, furans) and fine particulate matter.

4. Global Warming and Ozone Depletion:

Release of **CFCs** and **HCFCs** from e-waste refrigeration units contributes to greenhouse gas effects and ozone layer damage.

5. Health Impacts of Hazardous E-Waste Components

- **Lead:** Damages central nervous system (CNS); particularly harmful to children.
- **Mercury:** Affects brain and kidneys; organic mercury compounds are extremely toxic.
- **Cadmium:** Causes kidney failure and skeletal damage (Itai-Itai disease).
- **Chromium VI:** Causes DNA damage and lung cancer.
- **Beryllium:** Inhalation causes chronic beryllium disease.
- **PBDEs/PCBs:** Endocrine disruption, developmental and reproductive toxicity.
- **PVC burning:** Produces dioxins – among the most toxic substances known.

6. Management and Mitigation Measures

1. Safe Recycling Practices:

- Use certified e-waste recycling facilities.
- Avoid manual dismantling and open burning.

2. Substitution and Design for Environment (DfE):

- Replace lead-based solders with lead-free alternatives (e.g., tin-silver-copper).
- Use halogen-free flame retardants.

3. Proper Collection and Segregation:

- Source segregation of hazardous components before recycling.

4. Extended Producer Responsibility (EPR):

- Producers should take responsibility for end-of-life management.

5. Regulations:

- **Basel Convention (1989):** Controls transboundary movement of hazardous waste.
- **E-Waste Management Rules, 2022 (India):** Mandates EPR and safe recycling.

Global RoHS Compliances (RoHS Directive 84)

1. Introduction

RoHS stands for Restriction of Hazardous Substances. It is one of the most significant environmental directives aimed at controlling hazardous substances in electrical and electronic equipment (EEE).

The original RoHS Directive 2002/95/EC, often referred to as the RoHS Directive (EU Directive 84) due to its relation to the European Union environmental framework (84/500/EEC onwards), came into effect in 2003 and was implemented in 2006 across the EU.

The main objective of RoHS is to reduce the environmental and health impact of hazardous substances present in electrical and electronic products, particularly during their manufacturing, recycling, and disposal phases.

2. Objectives of RoHS Directive

1. **To minimize hazardous substances** in electrical and electronic equipment (EEE).
2. **To promote recycling and recovery** of e-waste by preventing toxic contamination.
3. **To ensure safer working conditions** for manufacturing and recycling workers.
4. **To align global standards** for product safety and environmental protection.
5. **To complement the WEEE Directive** (Waste Electrical and Electronic Equipment Directive) by ensuring that products entering the waste stream are free from dangerous materials.

3. Key Features of the RoHS Directive

1. Scope:

Applies to most categories of EEE — including large household appliances, IT and telecom equipment, consumer electronics, lighting equipment, tools, toys, and automatic dispensers.

2. Prohibition:

Prohibits the use of specific hazardous materials above set threshold levels in EEE sold in EU markets.

3. Manufacturer Responsibility:

Manufacturers must ensure compliance through material declaration, testing, and documentation before selling products.

4. Compliance Proof:

Requires **CE marking** on products to indicate conformity with RoHS requirements.

4. Restricted Substances under RoHS Directive

Originally, **six hazardous substances** were restricted under **RoHS 1 (2002/95/EC)**. Later, **RoHS 2 (2011/65/EU)** and **RoHS 3 (2015/863/EU)** added four more.

No.	Substance	Maximum Limit (by weight in homogeneous material)	Health/Environmental Hazards
1	Lead (Pb)	0.1%	Neurotoxic, developmental disorders
2	Mercury (Hg)	0.1%	Damages brain and kidneys
3	Cadmium (Cd)	0.01%	Carcinogenic, kidney damage
4	Hexavalent Chromium (Cr ⁶⁺)	0.1%	Carcinogenic, respiratory problems
5	Polybrominated Biphenyls (PBBs)	0.1%	Persistent organic pollutants
6	Polybrominated Diphenyl Ethers (PBDEs)	0.1%	Endocrine disruptors
7	Bis(2-ethylhexyl) phthalate (DEHP)	0.1%	Reproductive toxicant
8	Benzyl butyl phthalate (BBP)	0.1%	Hormonal interference
9	Dibutyl phthalate (DBP)	0.1%	Affects fertility
10	Diisobutyl phthalate (DIBP)	0.1%	Developmental toxicant

5. Categories of EEE Covered under RoHS

1. Large and small household appliances
2. IT and telecommunications equipment
3. Consumer electronics (e.g., TVs, radios)
4. Lighting equipment (lamps, bulbs, LEDs)
5. Electrical and electronic tools
6. Toys, leisure, and sports equipment
7. Automatic dispensers
8. Monitoring and control instruments (industrial and medical)

6. Exemptions under RoHS

Certain applications are exempted due to lack of feasible alternatives:

- Lead in high-temperature solders (>85% Pb)
- Mercury in specific medical or calibration devices
- Cadmium in photovoltaic panels
- Hexavalent chromium in certain corrosion-resistant coatings

Exemptions are reviewed periodically as technology advances.

7. Global Implementation of RoHS (RoHS-like Regulations)

The RoHS Directive originated in the **European Union**, but its influence is now **global**. Many countries have adopted similar regulations, often referred to as “**RoHS-like**” directives.

Region / Country	Regulation Name	Implementation Year	Remarks
European Union	RoHS 1 (2002/95/EC),	2006–2019	Basis for global standard
	RoHS 2 (2011/65/EU),		
	RoHS 3 (2015/863/EU)		
	)		
China	China RoHS (Management Methods for	2006	Similar to EU RoHS; focuses on

Region / Country	Regulation Name	Implementation Year	Remarks
	the Control of Pollution from Electronic Information Products)		labeling and disclosure
Japan	J-MOSS (Japan RoHS)	2006	Emphasizes product marking and consumer awareness
USA (various states)	California RoHS, New Jersey RoHS	2007 onwards	State-level adoption; aligns with EU limits
India	E-Waste Management Rules, 2016 (amended 2022)	2016	RoHS provisions under Rule 16; restricts six substances
Korea	K-RoHS	2008	Integrated under “Act for Resource Recycling of Electrical/Electronic Products”
Singapore	Environmental Protection and Management (RoHS) Regulations	2016	Aligns closely with EU RoHS 2
Turkey, UAE, Brazil, Canada	National RoHS frameworks	2015–2020	Adopted to ensure trade compatibility and environmental safety

8. RoHS Compliance Process

To ensure that products meet RoHS requirements, manufacturers and importers must:

1. **Identify Restricted Substances:** Review Bill of Materials (BOM) for compliance.

2. **Testing and Verification:** Use X-Ray Fluorescence (XRF) and chemical analysis for detection.
3. **Documentation and Declaration:** Prepare a Declaration of Conformity (DoC) and Technical File for inspection.
4. **Product Labelling:** Apply **CE Marking** to products indicating RoHS compliance.
5. **Periodic Audits:** Conduct regular supplier audits and testing to ensure ongoing compliance.

9. Relationship between RoHS and WEEE Directives

Aspect	RoHS Directive	WEEE Directive
Focus	Restricts hazardous substances in new products	Manages collection, recycling, and recovery of e-waste
Objective	Prevent pollution at the design stage	Manage waste after product use
Implementation Stage	Before product manufacturing and sale	After product becomes waste
Connection	Reduces hazardous waste in WEEE stream	Depends on RoHS to ensure safer waste composition

10. Benefits of RoHS Compliance

- Reduces environmental contamination.
- Protects human health, especially workers in recycling industries.
- Promotes green design and cleaner production.
- Facilitates international trade compliance.
- Supports circular economy principles and sustainable electronics.

11. Challenges in RoHS Implementation

- High cost of material testing and certification.
- Supply chain complexity and lack of transparency.
- Technical limitations for certain components.
- Differences between national RoHS laws (EU vs China vs India).

RoHS Compliance Requirements in India

1. Introduction

RoHS stands for Restriction of Hazardous Substances. It is an environmental regulation designed to limit or eliminate the use of certain hazardous materials in the manufacture of Electrical and Electronic Equipment (EEE).

In India, RoHS compliance is an integral part of the E-Waste (Management) Rules, 2016, which were notified under the Environment (Protection) Act, 1986. These rules are aligned with the European Union's RoHS Directive (2011/65/EU) to ensure environmental safety and global trade compatibility.

The main purpose of RoHS in India is to reduce toxic substances in electronic products, making them safer for human health and the environment during use, recycling, and disposal.

2. Legislative Background

1. **Environment (Protection) Act, 1986** – The parent act empowering the Central Government to regulate environmental pollution.
2. **E-Waste (Management) Rules, 2016** – These rules replaced the 2011 version and included RoHS provisions in **Rule 16**.
3. **Amendments in 2018 & 2022** – Strengthened compliance mechanisms, expanded coverage, and clarified producer responsibilities.

3. Objective of RoHS Compliance in India

- To minimize the presence of hazardous substances in electronic products.
- To ensure eco-friendly design and promote “Green Electronics.”
- To reduce pollution during e-waste recycling and disposal.
- To encourage Extended Producer Responsibility (EPR) by making producers accountable for compliance.
- To align Indian standards with international environmental norms (EU RoHS, China RoHS, etc.).

4. Scope of Application

The RoHS provisions in India apply to **Electrical and Electronic Equipment (EEE)** placed in the Indian market under **Schedule I** of the E-Waste Management Rules. It includes 21 categories of products such as:

1. Large and small household appliances
2. IT and telecommunication equipment
3. Consumer electronics (TVs, audio systems, etc.)
4. Lighting equipment
5. Electrical and electronic tools
6. Toys and sports equipment
7. Monitoring and control instruments
8. Automatic dispensers
9. Medical devices (with some exemptions)

5. RoHS Compliance Requirements for Producers

As per the E-Waste (Management) Rules, 2016 (Rule 16), every producer, manufacturer, and importer of EEE must:

(a) Limit Use of Hazardous Substances

Ensure that new equipment does not contain restricted substances beyond the specified limits.

(b) Declaration of Conformity

- Producers must issue a self-declaration confirming compliance with RoHS limits.
- This declaration must be submitted to the **Central Pollution Control Board (CPCB)**.
- Labels on products must mention “**Complies with the E-Waste (Management) Rules, 2016 – Restriction of Hazardous Substances**”.

(c) Technical Documentation

Maintain documentation including:

- Material composition reports
- Supplier declarations
- Test reports from accredited labs

These documents serve as proof during audits or inspections by pollution control authorities.

(d) Extended Producer Responsibility (EPR)

Producers are responsible for:

- Establishing collection centres or take-back systems.
- Ensuring environmentally sound recycling of e-waste.

- Submitting EPR plans and targets to CPCB for approval.

(e) Record Keeping and Audits

Maintain records of products manufactured, imported, or sold. The CPCB may conduct periodic audits or require sample testing for verification.

6. Testing and Verification of Compliance

- Testing must be done using recognized laboratories accredited by NABL (National Accreditation Board for Testing and Calibration Laboratories).
- Analytical methods such as X-Ray Fluorescence (XRF) or Inductively Coupled Plasma (ICP) spectroscopy are used.
- Manufacturers must provide a Material Composition Report (MCR) or Test Certificate to CPCB.

7. Exemptions under Indian RoHS

Certain applications are temporarily exempted due to technical or economic infeasibility:

Exemption Example	Description
Lead in high-temperature solder alloys	Used in servers, network infrastructure
Mercury in certain fluorescent lamps	Until suitable alternatives are available
Cadmium in solar panels	Limited exemption for photovoltaic applications
Hexavalent chromium in corrosion coatings	For heavy-duty industrial use

The **MoEFCC (Ministry of Environment, Forest and Climate Change)** periodically reviews and updates these exemptions.

8. Regulatory and Enforcement Agencies

Authority	Role
MoEFCC	Policy formulation and overall coordination
Central Pollution Control Board (CPCB)	Monitoring, certification, and enforcement

Authority	Role
State Pollution Control Boards (SPCBs)	Regional implementation and inspections
BIS (Bureau of Indian Standards)	Aligns product standards with RoHS requirements

9. Penalties for Non-Compliance

Failure to comply with RoHS norms may result in:

- Revocation of authorization or EPR registration
- Monetary penalties under the Environment (Protection) Act, 1986
- Product recall or ban on sale
- Legal prosecution for serious violations

These measures ensure accountability among manufacturers and importers.

10. Benefits of RoHS Compliance in India

1. Reduces environmental pollution and toxic e-waste.
2. Protects worker health in recycling and manufacturing sectors.
3. Encourages innovation and sustainable product design.
4. Enhances India's export competitiveness (especially to EU and US markets).
5. Supports national initiatives like “**Make in India**” and “**Digital India**” with a green focus.

11. Challenges in Implementation

- Lack of awareness among small-scale manufacturers.
- Limited number of accredited testing labs.
- Difficulty in verifying compliance across large supply chains.
- High cost of RoHS-compliant materials.
- Enforcement challenges in the unorganized sector.

12. Recent Developments (2022 Amendment Highlights)

- Strengthened EPR portal system for registration and tracking.

- Mandatory online filing of Annual Returns for producers.
- Encouragement for authorized recyclers and formal sector recycling.
- Proposal to include additional hazardous substances (e.g., phthalates) in future updates.

E-Waste Recycling: E-Waste Recycling Operations

1. Introduction

E-waste recycling refers to the collection, dismantling, separation, and recovery of valuable materials from discarded electrical and electronic equipment (EEE). Since e-waste contains both valuable materials (like gold, copper, aluminum, plastics) and hazardous substances (like lead, mercury, cadmium, and brominated compounds), recycling must be done through environmentally sound and safe operations.

The main goal of e-waste recycling is to minimize waste going to landfills, recover useful resources, and prevent environmental pollution.

2. Objectives of E-Waste Recycling Operations

- To recover valuable metals (gold, silver, copper, aluminum, etc.) for reuse.
- To reduce environmental contamination from hazardous substances.
- To conserve natural resources by substituting primary (mined) materials with secondary (recycled) materials.
- To promote circular economy through reuse, refurbishing, and recycling.
- To create green jobs in formal recycling sectors.

3. Overview of E-Waste Recycling Process

The e-waste recycling process generally involves the following major stages:

1. Collection and Transportation
2. Storage and Sorting
3. Dismantling (Manual or Mechanical)
4. Mechanical Processing (Size Reduction & Separation)
5. Material Recovery (Metals, Plastics, Glass)
6. Refining and Smelting

7. Treatment and Disposal of Residues

Each of these stages is discussed below in detail.

4. Stages of E-Waste Recycling Operations

4.1 Collection and Transportation

- **Collection:**
 - E-waste is collected from households, offices, retailers, repair shops, and bulk consumers.
 - Collection centres are often operated by producers (under EPR) or authorized recyclers.
 - Example: Drop-off points, take-back systems, and e-waste bins.
- **Transportation:**
 - Transported to authorized dismantlers or recyclers using vehicles that comply with hazardous waste transportation norms (with labelling and manifests).

4.2 Storage and Sorting

- Collected waste is stored and sorted based on:
 - Product category (computers, TVs, mobiles, etc.)
 - Reusability (working vs non-working)
 - Type of material (plastic, metal, PCB, etc.)
- Reusable devices may be refurbished or donated.
- Sorting helps in efficient downstream processing and maximizes material recovery.

4.3 Dismantling (Manual or Semi-Automatic)

- This is a crucial step in recycling operations.
- Workers manually dismantle devices to separate components such as:
 - Printed Circuit Boards (PCBs)
 - CRTs / Displays
 - Cables and wiring
 - Batteries
 - Plastic and metal housings

- Hard drives and processors
- Tools used: screwdrivers, cutters, de-soldering stations.
- Dismantling helps to segregate hazardous components for safe treatment.

Safety Note: Workers must use PPE (Personal Protective Equipment) like gloves, masks, and safety glasses due to toxic dust exposure.

4.4 Mechanical Processing (Size Reduction and Separation)

After dismantling, mechanical operations are carried out to **crush, shred, and separate** different materials.

1. Shredding:

- Large e-waste components are broken down into smaller pieces.
- Equipment: shredders, hammer mills, granulators.

2. Magnetic Separation:

- Used to separate **ferrous metals (iron, steel)** using magnets.

3. Eddy Current Separation:

- Separates **non-ferrous metals** like aluminium and copper from plastics.

4. Density Separation (Air or Water Table):

- Uses air or water flow to separate materials based on density differences.

5. Optical Sorting:

- Sensors detect and separate materials by colour and type (used for plastics).

These mechanical processes help in obtaining **metal-rich fractions** and **non-metal fractions** for further refining.

4.5 Material Recovery (Metals, Plastics, Glass)

After separation, each material stream is processed to recover pure forms:

(a) Metal Recovery

- Metal-rich fractions from PCBs are sent to **pyrometallurgical** or **hydrometallurgical** refineries.
- **Commonly recovered metals:** Copper, gold, silver, palladium, aluminium, tin, and nickel.

(b) Plastic Recovery

- Plastics are cleaned, shredded, and melted for reuse in new products.

- Non-recyclable plastics containing **brominated flame retardants (BFRs)** are treated in **specialized incinerators**.

(c) Glass Recovery

- CRT glass and display panels are treated separately.
- Leaded glass (from CRTs) is reused in the battery or lead smelting industry.

4.6 Refining and Smelting

Metallurgical operations are used to extract precious metals and refine base metals.

Method	Description	Example Recovered Metals
Pyrometallurgy	High-temperature smelting of PCBs and metal fractions	Cu, Au, Ag, Pb
Hydrometallurgy	Leaching metals using chemical solutions (acid or cyanide-based)	Au, Ag, Cu
Electrometallurgy	Electrolysis-based recovery of metals from leachate	Cu, Ni, Zn

These processes must include **pollution control units** to handle fumes, dust, and residues.

4.7 Treatment and Disposal of Residuals

- Non-recyclable materials, toxic residues, and sludge are disposed of at authorized Treatment, Storage, and Disposal Facilities (TSDFs).
- Hazardous fractions (e.g., mercury lamps, batteries) are treated using stabilization, encapsulation, or secure landfilling methods.

5. Flow Diagram of E-Waste Recycling Process

Collection → Transportation → Sorting → Dismantling →

Mechanical Separation → Metal/Plastic Recovery → Refining → Disposal

6. Advanced and Emerging Recycling Technologies

Technology	Description
Bioleaching	Use of microorganisms to extract metals from e-waste (eco-friendly method).

Technology	Description
Cryogenic Processing	Freezing materials for easy cracking and separation.
Plasma Arc Recycling	High-temperature plasma used to recover metals from complex waste.
Automated Dismantling	Robotic AI/robotic systems to safely dismantle devices (used in Japan, EU).

These emerging technologies aim to increase recovery efficiency and reduce environmental impact.

7. Environmental and Health Safety in Recycling Operations

- Workers must be provided with **PPE, ventilation systems, and training**.
- Recyclers must have:
 - Dust collection units
 - Effluent Treatment Plants (ETP)
 - Air pollution control systems
- Proper waste handling prevents:
 - Air contamination (from burning plastics)
 - Water pollution (from acid leaching)
 - Health risks (from heavy metal exposure)

8. Role of Authorized Recyclers in India

Under the E-Waste (Management) Rules, 2016, only CPCB-registered and authorized recyclers can process e waste. They are responsible for:

- Ensuring **safe dismantling and material recovery**.
- Maintaining **records** and submitting annual returns to CPCB/SPCB.
- Operating within approved environmental and safety standards.

Examples: **Attero Recycling (Noida), TES-AMM (Bangalore), E-Parisaraa (Bangalore)**.

9. Benefits of E-Waste Recycling Operations

- Reduces environmental pollution.
- Recovers valuable metals and materials.

- Reduces demand for mining and raw material extraction.
- Generates employment in formal recycling sectors.
- Supports circular economy and sustainable development goals (SDGs).

10. Challenges in E-Waste Recycling

- Dominance of the informal sector (unregulated backyard recycling).
- Lack of public awareness and proper collection systems.
- High cost of advanced recycling technology.
- Weak enforcement of RoHS and EPR provisions.

Dismantling and Segregation in E-Waste Recycling

1. Introduction

E-waste dismantling and segregation are critical stages in the e-waste recycling process. They form the initial steps of material recovery, where complex electronic products are taken apart, and their components are separated into different material streams (metals, plastics, glass, etc.) for further processing.

E-waste dismantling not only helps in recovering valuable materials but also ensures safe handling of hazardous components, reducing environmental and human health risks.

2. Objectives of Dismantling and Segregation

1. To separate reusable, recyclable, and hazardous components from mixed e-waste.
2. To prepare e-waste for downstream recycling processes (mechanical, thermal, or chemical recovery).
3. To maximize recovery efficiency by improving material purity.
4. To prevent contamination and enable safe disposal of toxic fractions.
5. To facilitate proper treatment of hazardous materials like lead, mercury, and cadmium.

3. Importance in the E-Waste Recycling Chain

Dismantling and segregation serve as a bridge between collection and mechanical/chemical recycling.

They ensure that:

- Reusable parts (like RAM, CPUs, power supplies) are recovered for refurbishing.
- Hazardous fractions (like batteries, CRTs, mercury lamps) are isolated safely.
- Material-specific streams (metals, plastics, glass) are prepared for processing.

Without proper dismantling and segregation, recycling efficiency drops and contamination risk increases.

4. Types of Dismantling Operations

E-waste dismantling can be **manual**, **semi-mechanical**, or **fully automated**, depending on the scale and technology used.

Type	Description	Used For
Manual Dismantling	Hand tools used to disassemble parts	Common in small and medium recyclers
Semi-Mechanical Dismantling	Combination of manual disassembly and machines (cutters, crushers)	Large-scale recyclers
Automated Dismantling	Robotic or AI-based systems for precision dismantling	Advanced facilities (e.g., Japan, EU)

5. Steps Involved in Dismantling and Segregation

Step 1: Receiving and Inspection

- E-waste received from collection centers is **inspected** for product type and condition.
- Devices that can be **reused or refurbished** are separated.

Step 2: De-pollution

- Removal of **hazardous or pollutant components** before mechanical processing. These include:
 - Batteries (Ni-Cd, Li-ion, lead-acid)
 - Mercury-containing switches, bulbs
 - Ink cartridges, toner
 - Capacitors with PCBs
 - CRTs (lead glass)

This step ensures safe handling and prevents toxic emissions in later stages.

Step 3: Manual Dismantling

- Products are **opened using simple hand tools** like screwdrivers, pliers, and desoldering guns.
- Components are **sorted and separated** into categories such as:
 - Printed Circuit Boards (PCBs)
 - Plastic casings
 - Cables and wiring
 - Ferrous and non-ferrous metal parts
 - Glass and CRT components
 - Hard drives, motors, and fans

Example:

In a desktop computer:

- The metal casing → sent for metal recycling
- PCB → sent for copper and gold recovery
- Plastic → sent for granulation
- CRT monitor → sent for lead glass treatment

Step 4: Segregation of Materials

After dismantling, materials are **segregated** based on their **composition** and **recyclability**:

Category	Examples	Processing Route
Ferrous Metals	Steel, iron parts	Magnetic separation → metal recycling
Non-Ferrous Metals	Copper, aluminum, brass	Eddy current separation → smelting/refining
Plastics	Casings, insulation	Cleaning → granulation → reuse
Glass	CRT screens, LCD panels	Lead recovery or reuse in glass industry
Electronic Components	PCBs, chips, capacitors	Precious metal recovery
Hazardous Waste	Batteries, mercury, PCBs	Authorized hazardous waste treatment

Step 5: Storage and Labeling

- Each segregated material is **stored separately** in labeled containers.
- Hazardous components are stored in **safe, ventilated, and leak-proof areas** until sent to authorized recyclers or disposal facilities.

6. Equipment Used in Dismantling and Segregation

- **Hand Tools:** Screwdrivers, cutters, pliers, wrenches
- **De-soldering Equipment:** Hot air guns, soldering irons
- **Mechanical Tools:** Shredders, magnetic separators, eddy current separators
- **Safety Equipment:** Gloves, goggles, masks, ventilation systems

7. Safety and Environmental Considerations

During dismantling, workers are often exposed to **dust, fumes, and toxic materials**. Hence, safety is critical.

Safety Measures:

- Use of **PPE** (gloves, goggles, masks, aprons).
- Proper **ventilation** and **dust extraction systems**.
- Avoidance of open burning or acid leaching.
- **Training** for workers on handling hazardous components.
- **Emergency procedures** for spills or injuries.

Environmental Measures:

- Maintain **separate zones** for hazardous and non-hazardous waste.
- Avoid contamination of soil or water through proper containment.
- Dispose of residues at **authorized Treatment, Storage, and Disposal Facilities (TSDFs)**.

8. Dismantling and Segregation Flow Chart

Collection → Inspection → De-pollution → Manual Dismantling →

Segregation (Metals, Plastics, Glass, Hazardous) → Storage → Recycling/Disposal

9. Role of Authorized Dismantlers in India

Under the **E-Waste (Management) Rules, 2016**, dismantling can only be performed by **authorized facilities** approved by the **CPCB (Central Pollution Control Board)** or **SPCB (State Pollution Control Board)**.

Authorized dismantlers are responsible for:

- Safe removal of components.

- Maintaining proper records and manifests.
- Sending recovered materials to registered recyclers only.
- Ensuring compliance with environmental standards.

Examples of Authorized Dismantlers in India:

- **E-Parisaraa Pvt. Ltd. (Bangalore)**
- **Attero Recycling (Noida)**
- **TES-AMM India (Bangalore)**

10. Challenges in Dismantling and Segregation

- 1. Informal Sector Dominance:**
 - Many dismantlers operate illegally without safety measures.
- 2. Lack of Advanced Equipment:**
 - Manual dismantling leads to inefficiencies.
- 3. Worker Health Risks:**
 - Exposure to lead dust, solder fumes, and broken glass.
- 4. Poor Segregation Practices:**
 - Leads to cross-contamination and lower material recovery rates.
- 5. Regulatory Gaps and Limited Awareness:**
 - Need for stricter enforcement and better training programs.

11. Advantages of Proper Dismantling and Segregation

- Increases recovery efficiency and resource value.
- Reduces pollution from improper disposal.
- Prevents health hazards to workers and environment.
- Facilitates recycling and circular economy.
- Enhances traceability and compliance under **EPR (Extended Producer Responsibility)**.

E-Waste Recycling and Recovery

1. Introduction

E-waste recycling and recovery are crucial stages in the management of electronic waste, aimed at retrieving valuable materials (like metals, plastics, and glass) and safely disposing of hazardous substances.

E-waste is a complex mixture of valuable resources (gold, silver, copper, aluminum, and rare earth elements) and toxic materials (lead, mercury, cadmium, and brominated flame retardants). Through scientific recycling and recovery operations, these materials are extracted in an environmentally safe and economically beneficial way.

2. Objectives of Recycling and Recovery

1. To **recover valuable materials** (metals, plastics, glass) from e-waste.
2. To **minimize environmental pollution** caused by improper disposal.
3. To **reduce demand for virgin raw materials** (mining and refining).
4. To **promote circular economy** by reusing recovered materials.
5. To **reduce landfill burden** and prevent leaching of toxic elements.

3. Difference Between Recycling and Recovery

Aspect	Recycling	Recovery
Definition	Process of converting waste materials into reusable products or raw materials.	Extraction of valuable substances (mainly metals) from e-waste through physical or chemical methods.
Focus	Material reprocessing (plastic, glass, metal).	Metal recovery (copper, gold, silver, etc.).
Techniques	Mechanical separation, melting, reprocessing.	Pyrometallurgical, hydrometallurgical, or biometallurgical extraction.
Output	Reusable raw materials (plastic pellets, glass cullets, metal ingots).	Refined metals, alloys, and reusable resources.

4. Stages of Recycling and Recovery Operations

E-waste recycling and recovery usually follow a **stepwise process** after dismantling and segregation.

Stage 1: Collection and Sorting

- E-waste is collected and sorted according to product category (computers, mobiles, TVs, etc.) and material type.
- Segregation helps direct specific components to appropriate recovery methods.

Stage 2: Dismantling

- Devices are manually or mechanically dismantled to remove hazardous and reusable parts:
 - Batteries, CRTs, PCBs, cables, plastic casings, etc.
- Prepares waste for further mechanical or metallurgical treatment.

Stage 3: Mechanical Recycling

Involves **physical size reduction and material separation**.

Key Processes:

1. **Shredding:**
 - Breaks e-waste into small pieces to expose materials.
2. **Magnetic Separation:**
 - Extracts **ferrous metals** (iron, steel) using magnets.
3. **Eddy Current Separation:**
 - Separates **non-ferrous metals** (aluminum, copper).
4. **Density or Air Classification:**
 - Separates light (plastic) and heavy (metal/glass) materials.
5. **Optical Sorting:**
 - Identifies and separates plastics by color or resin type.

Outputs:

- Metal-rich fraction → sent for metal recovery.
- Plastic fraction → sent for reprocessing.
- Glass → sent to glass recycling units.

Stage 4: Material-Specific Recycling

(a) Plastic Recycling

- Plastics from casings, keyboards, and cables are:
 - Cleaned and shredded.
 - Melted and extruded into pellets.
- Non-recyclable plastics containing **brominated flame retardants (BFRs)** are treated via **co-processing in cement kilns** or **incineration with flue gas treatment**.

(b) Glass Recycling

- Glass from CRTs and screens is separated:
 - **Panel glass** (low lead) → reused in new screens or glass products.
 - **Funnel glass** (high lead) → sent for lead recovery or safe disposal.

(c) Metal Recycling

- Basic metals (iron, copper, aluminium) are melted in furnaces and purified for reuse in manufacturing industries.

Stage 5: Metal Recovery from Printed Circuit Boards (PCBs)

PCBs are rich in valuable and precious metals like **copper, gold, silver, platinum, and palladium**.

Recovery from PCBs is the most **technically intensive** step in e-waste recycling.

Stage 6: Refining

- Metals recovered are refined to improve purity (e.g., electrorefining for copper and gold).
- Recovered metals can then be used in **electronics, jewelry, or industrial manufacturing**.

Stage 7: Disposal of Residual Waste

- Non-recyclable residues, sludges, and contaminated fractions are disposed of at **authorized Treatment, Storage, and Disposal Facilities (TSDFs)**.
- Ensures no leaching of heavy metals into soil or water.

5. Recycling and Recovery Flow Diagram

Collection → Dismantling → Mechanical Recycling →

Material Separation → Metal Recovery → Refining → Safe Disposal

6. Technologies Used in E-Waste Recycling and Recovery

Category	Examples of Technology	Application
Mechanical	Shredders, magnetic separators, eddy current separators	Size reduction and material sorting
Chemical	Acid leaching, cyanide process	Precious metal recovery
Thermal	Pyrolysis, plasma arc melting	Metal extraction and plastic treatment
Biological	Bioleaching using bacteria	Metal extraction in eco-friendly way

7. Safety and Environmental Considerations

Recycling and recovery must be carried out in **authorized, pollution-controlled facilities** with proper safety measures:

- **Personal Protective Equipment (PPE)** for workers.
- **Effluent Treatment Plants (ETPs)** for liquid waste.
- **Air Pollution Control Devices (APCDs)** for fumes.
- **Proper ventilation and dust control systems.**
- **No open burning or acid leaching** in the informal sector.

8. Role of Authorized Recyclers in India

Under the **E-Waste (Management) Rules, 2016**, recycling and recovery can be performed **only by registered recyclers authorized by the Central Pollution Control Board (CPCB).**

Authorized recyclers:

- Follow environmentally sound recycling methods.
- Maintain **records of input/output** and submit **annual returns.**
- Handle hazardous residues safely.
- Work under **Extended Producer Responsibility (EPR)** framework.

Examples in India:

- **Attero Recycling (Noida)** – precious metal recovery.

- **E-Parisaraa Pvt. Ltd. (Bangalore)** – mechanical recycling.
- **TES-AMM India (Bangalore)** – global-grade electronic recovery.

9. Benefits of Recycling and Recovery

1. **Environmental Protection:** Prevents soil and water contamination.
2. **Resource Conservation:** Recovers valuable metals and materials.
3. **Energy Saving:** Recycling uses less energy than mining raw materials.
4. **Economic Gains:** Generates income from recovered metals and plastics.
5. **Employment Opportunities:** Creates green jobs in the formal recycling sector.
6. **Supports Circular Economy:** Promotes sustainable production and consumption.

10. Challenges in Recycling and Recovery

- **Dominance of the informal sector** (unsafe recycling practices).
- **Lack of advanced technologies** and infrastructure.
- **Complex material composition** of modern electronics.
- **High cost** of chemical and metallurgical recovery.
- **Regulatory enforcement gaps** in developing countries.

11. Future Trends in E-Waste Recycling and Recovery

- **Urban Mining:** Extracting metals from discarded electronics as a secondary source.
- **AI and Robotics:** Automated dismantling and sorting.
- **Green Chemistry:** Non-toxic leaching agents for metal recovery.
- **Circular Product Design:** Design-for-recycling approaches by manufacturers.
- **Integration of Informal Sector:** Formal training and licensing programs.

Recycling Technologies in E-Waste Management

1. Introduction

Electronic waste (e-waste) consists of discarded electrical and electronic equipment such as computers, mobile phones, televisions, and household appliances. These items contain a wide variety of materials, including metals, plastics, glass, and ceramics, as well as hazardous substances like lead, mercury, and cadmium.

Recycling technologies are the methods and systems used to recover valuable materials from e-waste while minimizing environmental pollution. These technologies can be mechanical, chemical, thermal, or biological, depending on the nature of the material and the recovery process.

The main goal of recycling technologies is to extract reusable materials, reduce hazardous impacts, and promote circular economy by reintegrating recovered materials into production chains.

2. Objectives of E-Waste Recycling Technologies

1. To **recover valuable metals** (Cu, Al, Fe, Au, Ag, Pd, etc.) from discarded electronics.
2. To **minimize environmental pollution** from hazardous components.
3. To **reduce the extraction of virgin raw materials** and conserve natural resources.
4. To **generate employment** in environmentally sound recycling sectors.
5. To **support Extended Producer Responsibility (EPR)** and sustainable waste management.

3. Classification of E-Waste Recycling Technologies

E-waste recycling involves a series of technologies applied at different stages. The main categories are:

Type of Technology	Main Principle	Examples / Applications
Mechanical Physical	/ Separation of materials based on physical properties	Shredding, magnetic separation, eddy current separation
Thermal	Use of heat to recover metals or degrade plastics	Pyrometallurgy, incineration, pyrolysis,

Type of Technology	Main Principle	Examples / Applications
Chemical	Dissolution and recovery of metals through chemical reactions	Hydrometallurgy, electrolysis, solvent extraction
Biological	Use of microorganisms to extract metals	Bioleaching, biosorption

4. Mechanical (Physical) Recycling Technologies

4.1. Overview

Mechanical processes involve **size reduction, separation, and classification** of e-waste materials based on physical characteristics like **density, magnetism, or conductivity**. These are usually **pre-treatment steps** before metallurgical recovery.

4.2. Steps Involved

1. Shredding

- E-waste is broken into small fragments (2–10 mm).
- Exposes metals, plastics, and ceramics for easier separation.

2. Magnetic Separation

- Uses magnets to extract **ferrous metals** (iron, steel).

3. Eddy Current Separation

- Separates **non-ferrous metals** (copper, aluminum, brass) using electromagnetic induction.

4. Density / Air Classification

- Separates light materials (plastics) from heavier ones (metals, glass).

5. Optical / Electrostatic Separation

- Advanced methods for sorting plastics and metals based on conductivity or color recognition.

4.3. Advantages

- Environmentally clean process (no chemicals used).
- Simple and cost-effective.
- Suitable for large-scale preliminary recycling.

4.4. Limitations

- Limited recovery efficiency for fine or composite materials.

- Requires further metallurgical processing to extract metals.

5. Thermal Recycling Technologies

5.1. Overview

Thermal technologies use **high-temperature treatments** to separate or recover valuable materials and destroy organic contaminants such as plastics and resins.

5.2. Major Thermal Techniques

(a) Pyrometallurgy

- Metals are recovered through **smelting** or **roasting** at high temperatures (up to 1200°C).
- Involves oxidation or reduction reactions to separate metals.

Example: Recovery of copper, gold, and silver from Printed Circuit Boards (PCBs).

Advantages:

- High recovery efficiency.
- Suitable for large volumes of mixed waste.

Disadvantages:

- High energy consumption.
- Emission of toxic gases (requires air pollution control).

(b) Pyrolysis

- Thermal decomposition of organic materials (plastics, resins) in **absence of oxygen**.
- Produces gases, oils, and solid residues containing metals.
- Helps recover metals embedded in plastic-based PCBs.

Example: Conversion of plastic waste into fuel oils.

Advantages:

- Reduces volume of waste.
- Recovers both metals and energy-rich oils.

Disadvantages:

- Requires temperature control and emission treatment.

(c) Incineration with Energy Recovery

- Controlled burning of e-waste to generate heat or electricity.

- Residual ash contains recoverable metals.

Advantages:

- Volume reduction up to 90%.
- Can generate electricity from waste heat.

Disadvantages:

- Not suitable for all e-waste types.
- Requires flue gas cleaning systems.

6. Chemical Recycling Technologies**6.1. Overview**

Chemical methods involve **leaching valuable metals** from e-waste using **acids, bases, or other chemical reagents**, followed by **precipitation, solvent extraction, or electrolysis** to recover pure metals.

6.2. Major Chemical Techniques**(a) Hydrometallurgy**

- Metals are dissolved from shredded e-waste using **aqueous solutions** (HCl, H₂SO₄, cyanide, thiourea, etc.).
- Followed by **selective precipitation or solvent extraction**.

Example:

Gold recovery using cyanide or thiourea leaching.

Advantages:

- High purity of recovered metals.
- Lower energy requirement compared to pyrometallurgy.

Disadvantages:

- Generates chemical effluents that require treatment.
- Slower and more complex than mechanical methods.

(b) Electrometallurgy (Electrolysis)

- Metal ions from leach solutions are recovered through **electro-deposition** on cathodes.
- Commonly used for **copper, nickel, and silver** recovery.

Advantages:

- Produces high-purity metals.

- Energy efficient when used with hydrometallurgy.

Disadvantages:

- Requires purified leach solutions.
- Limited to specific metals.

(c) Solvent Extraction

- Selective extraction of metals from mixed leach solutions using organic solvents.
- Used for separation of **rare earth elements (REEs)**.

Example: Extraction of neodymium from magnet waste.

7. Biological Recycling Technologies**7.1. Overview**

Bio-recycling uses **microorganisms (bacteria or fungi)** to recover metals from e-waste through **bioleaching** or **biosorption**.

7.2. Techniques**(a) Bioleaching**

- Microbes like *Acidithiobacillus ferrooxidans* and *Leptospirillum ferrooxidans* oxidize sulfide minerals and release metal ions into solution.

Metals Recovered: Copper, zinc, nickel, and gold.

Advantages:

- Eco-friendly and low energy consumption.
- No toxic chemicals required.

Disadvantages:

- Slow process (takes weeks to months).
- Requires controlled temperature and pH.

(b) Biosorption

- Microbial biomass or plant material adsorbs metal ions from waste streams.
- Suitable for trace metal recovery.

8. Emerging and Advanced Recycling Technologies

Technology	Principle / Description	Application
Plasma Arc Recycling	Uses extremely high temperatures (~5000°C) to melt and vaporize e-waste, leaving clean metal slag.	Treatment of complex waste streams.
Supercritical Fluid Extraction (SCFE)	Uses supercritical CO ₂ or water for metal and plastic recovery.	Environmentally friendly extraction.
Cryogenic Separation	Freezing e-waste to make materials brittle for easier separation.	Recycling of flexible circuit boards.
AI and Robotics in Dismantling	Automated sorting and dismantling using AI and sensors.	Efficient and precise material recovery.

9. Comparison of Different Recycling Technologies

Aspect	Mechanical	Thermal	Chemical	Biological
Energy Use	Low	High	Moderate	Very Low
Pollution Risk	Low	High (fumes)	Moderate (effluents)	Minimal
Recovery Efficiency	Moderate	High	Very High	Moderate
Cost	Low	High	High	Low
Best For	Initial processing	Large-scale metal recovery	Precious metals	Low-grade ores and eco-friendly processes

10. Environmental and Safety Considerations

- Use of **air pollution control systems (APCDs)** in thermal recycling.
- **Effluent treatment plants (ETPs)** for chemical residues.
- **Personal protective equipment (PPE)** for workers.
- **Waste minimization and residue disposal** at authorized TSDFs.
- Strict compliance with **E-Waste (Management) Rules, 2016 and Amendments, 2022**.

11. Challenges in Implementing Recycling Technologies

1. High investment and operational costs.

2. Lack of advanced technology infrastructure in developing countries.
3. Presence of informal recycling sector using unsafe methods.
4. Complex composition of modern electronics.
5. Need for skilled workforce and technology transfer.

12. Future Prospects

- Development of **green leaching agents** (biodegradable and non-toxic).
- Integration of **AI-based sorting systems**.
- Growth of **urban mining** for precious metals.
- Increasing focus on **modular and design-for-recycling** electronics.
- Promotion of **Public-Private Partnerships (PPP)** for formal recycling units.